

Submission B solution to Problem 2

Theorem 1 (Headline theorem: original problem). Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let

$$\Sigma = \mathbb{R} \times [0, 1], \quad \partial\Sigma = \mathbb{R} \times \{0, 1\}.$$

For $\beta > 0$, define $G_\beta : \Sigma \rightarrow \Sigma$ by

$$G_\beta(x, y) = (x + \beta, 1 - y).$$

Let G_β^k denote the k -fold composition of G_β or G_β^{-1} , according as $k > 0$ or $k < 0$, and let G_β^0 be the identity. A *clean triangle* is the convex hull of three non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable, locally finite collection of clean triangles with pairwise disjoint interiors and union Σ . It is G_β -*invariant* if G_β preserves the triangulation. A *squeeze map* of a clean triangle is an affine map into $\mathbb{R}^3$ which decreases the length of the edge contained in $\partial\Sigma$ and increases the lengths of the other two edges. A number $\beta > 0$ is *realized* if there is a continuous map $f : \Sigma \rightarrow \mathbb{R}^3$ such that

1. f restricts to a squeeze map on each triangle of a G_β -invariant clean triangulation of Σ ;
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$.

Then the infimum of the set of realized numbers is $\sqrt{3}$.

Theorem 2 (Established stronger claim). *The realized numbers are exactly the interval $(\sqrt{3}, \infty)$.*

Overview / Strategy

The lower bound passes to the quotient $M_\beta = \Sigma / \langle G_\beta \rangle$, a flat Möbius band with boundary length 2β . The map descends to an embedded piecewise-affine surface. Inside each clean triangle the cevians from the apex to the boundary edge are strictly lengthened. These cevians form a leaf graph; in the quotient the leaf graph is a circle, while upstairs it is a line, and the deck transformation reverses the orientation of the image support line. A topological lemma about paths of oriented lines then forces two image support lines to be perpendicular and intersecting. Boundary shortening and a Euclidean estimate give $\beta > \sqrt{3}$.

For the upper bound we use one classical external input: the Sadowsky–Wunderlich construction of embedded developable Möbius bands, in a quantitative form. We approximate such a smooth ruled band by an invariant piecewise-linear ruled band whose boundary edges are not expanded and whose transverse edges are shortened by an arbitrarily small amount. A final anisotropic rescaling makes the boundary edges strictly shorter and all transverse edges strictly longer, yielding a realization for every $\beta > \sqrt{3}$.

1 The lower bound

Throughout this section assume that β is realized by f on a G_β -invariant clean triangulation $\mathcal{T}$. Write $G = G_\beta$.

Lemma 3 (Face-to-face structure). *The triangulation is face-to-face: distinct edges have disjoint relative interiors, every transverse edge is incident to exactly two triangles, and every boundary edge is incident to exactly one triangle.*

Proof. A clean triangle has exactly one boundary edge: among its three boundary vertices, exactly two lie on one boundary component and the third lies on the other. The remaining two edges are transverse.

Two transverse edges cannot cross in the interior of Σ , since near a transverse crossing each of the two incident triangles occupies one open side of its edge, and two such open half-disks always overlap in an open sector. If two transverse edges are collinear and overlap in more than one point, then they lie on the same non-horizontal line; such a line meets $y = 0$ and $y = 1$ once each, so the two edges are the same geometric segment. Boundary edges cannot overlap in a nontrivial interval, because at an interior point of the overlap both adjacent triangles contain an inward half-neighborhood of the boundary. Finally, the endpoint of a transverse edge cannot lie in the relative interior of a boundary edge: the triangle with the boundary edge contains a small inward half-disk there, while the triangle with the transverse edge contains a nonempty sector inside that half-disk. Hence distinct triangles meet only in common faces.

Let e be a transverse edge and z an interior point of e . A small disk around z is divided by e into two half-disks. At most one triangle occupies each side, by disjointness of interiors, and at least one triangle occupies each side, because the triangles cover Σ . Thus e is incident to exactly two triangles. The same half-disk argument at an interior point of a boundary edge gives exactly one incident triangle. $\square$

Lemma 4 (The finite quotient triangulation). *The quotient map f descends to an embedding $F : M_\beta \rightarrow \mathbb{R}^3$, where*

$$M_\beta = \Sigma / \langle G \rangle = [0, \beta] \times [0, 1] / (0, y) \sim (\beta, 1 - y),$$

and the quotient triangulation of M_β is finite and face-to-face.

Proof. Since $p = Gq$ implies $f(p) = f(q)$, the map descends to a continuous F . If $F([p]) = F([q])$, then $f(p) = f(q)$, so by realization $p = G^k q$, hence $[p] = [q]$. Thus F is injective; because M_β is compact and $\mathbb{R}^3$ is Hausdorff, F is an embedding.

The restriction of f to each triangle is injective. Indeed, if an affine restriction had rank < 2 , then arbitrarily close distinct points p, q in that triangle would satisfy $f(p) = f(q)$. Then $p = G^k q$ for some $k \neq 0$, impossible for sufficiently close p, q , since G^{2n} translates by $2n\beta$ and G^{2n+1} changes the x -coordinate by $(2n+1)\beta$, so every nontrivial G^k moves every point by distance at least β .

Therefore no closed triangle meets a nontrivial translate of itself. If $z \in \tau \cap G^k \tau$, write $z = G^k w$ with $w \in \tau$. Then $f(z) = f(w)$, and injectivity on τ gives $z = w$, contradicting freeness of the G -action unless $k = 0$. A compact fundamental rectangle, for instance $[0, 2\beta] \times [0, 1]$, meets only finitely many closed triangles by local finiteness and compactness, and every orbit of triangles has a representative meeting it. Hence the quotient triangulation is finite. Lemma 3 descends to the quotient. $\square$

Lemma 5 (Cevian lengthening). *Let $\Delta = \text{conv}\{a, r_0, r_1\}$ be a clean triangle whose boundary edge is $[r_0, r_1]$, and let $A : \Delta \rightarrow \mathbb{R}^3$ be a squeeze map. For $r(s) = (1-s)r_0 + sr_1$, every cevian $[a, r(s)]$ is strictly lengthened:*

$$|A(a) - A(r(s))| > |a - r(s)|, \quad 0 \leq s \leq 1.$$

Proof. Set $q(s) = |A(a) - A(r(s))|^2 - |a - r(s)|^2$. The two transverse edges are strictly lengthened, so $q(0) > 0$ and $q(1) > 0$. Since A is affine,

$$q''(s) = 2(|A(r_1) - A(r_0)|^2 - |r_1 - r_0|^2) < 0$$

because the boundary edge is strictly shortened. Thus q is strictly concave and lies above its chord: $q(s) \geq (1-s)q(0) + sq(1) > 0$. Taking square roots gives the claim. $\square$

Lemma 6 (Paths of oriented lines). *Let L_s , $0 \leq s \leq 1$, be a continuous path of oriented lines in $\mathbb{R}^3$. Suppose L_1 is the same unoriented line as L_0 , but with the opposite orientation. Then two distinct parameters give unoriented lines that are perpendicular and intersect.*

Proof. Represent an oriented line by (u, m) , where $u \in S^2$ is its unit direction and $m = p \times u$ for any point p on the line. Reversing orientation sends (u, m) to $(-u, -m)$. Choose continuous $(u(s), m(s))$ and extend by $(u(s+1), m(s+1)) = -(u(s), m(s))$. For $s \in \mathbb{R}/\mathbb{Z}$ and $0 \leq \delta \leq 1$, define

$$a(s, \delta) = u(s) \cdot u(s + \delta), \quad b(s, \delta) = u(s) \cdot m(s + \delta) + u(s + \delta) \cdot m(s).$$

Then $(a, b)(s, 0) = (1, 0)$, $(a, b)(s, 1) = (-1, 0)$, and $(a, b)(s + \delta, 1 - \delta) = -(a, b)(s, \delta)$.

Assume $(a, b) \neq (0, 0)$ everywhere. Then $\Phi = (a, b)/\sqrt{a^2 + b^2} : S^1 \times [0, 1] \rightarrow S^1$ is continuous and is constant on $\delta = 0$. On $\delta = 1/2$,

$$\Phi(s + 1/2, 1/2) = -\Phi(s, 1/2).$$

If $\psi : S^1 \rightarrow S^1$ has $\psi(s + 1/2) = -\psi(s)$, then, after lifting to an angle θ , the integer $(\theta(s + 1/2) - \theta(s) - \pi)/(2\pi)$ is constant, so the degree of ψ is odd. But $\Phi|_{\delta=1/2}$ is homotopic through $S^1 \times [0, 1/2]$ to the constant map $\Phi|_{\delta=0}$, so its degree is zero, a contradiction. Hence $(a, b) = (0, 0)$ somewhere, with $0 < \delta < 1$, so the two parameters are distinct.

At such a zero, $u \cdot v = 0$ for the two directions u, v . If the lines are represented by $(u, p \times u)$ and $(v, q \times v)$, then

$$b = u \cdot (q \times v) + v \cdot (p \times u) = (p - q) \cdot (u \times v).$$

Thus, when $u \perp v$, the condition $b = 0$ says the two nonparallel lines are coplanar. Nonparallel coplanar lines intersect. $\square$

Lemma 7 (Perpendicular intersecting leaves). *In M_β there are two distinct leaves whose image support lines under F are perpendicular and intersect.*

Proof. Inside each triangle take the cevians from the apex to the boundary edge and call them leaves. In the finite quotient triangulation, form a graph Λ_M by taking one closed interval for the leaves of each triangle and gluing endpoints corresponding to common transverse edges. By Lemma 3, every vertex has valence 2.

The graph is connected. If C is a component and $U_C \subset M_\beta$ is the union of the corresponding leaves, then U_C is open: this is immediate inside triangles, along boundary edges, and across transverse edges because the two incident leaf intervals are glued. At a boundary vertex, a small half-disk meets finitely many incident triangle sectors; consecutive sectors share an edge from the vertex, and all non-boundary shared edges are transverse, so all incident triangles lie in the same component. Thus the sets U_C are disjoint open subsets covering the connected surface M_β ; hence there is only one. A finite connected graph with all vertices of valence 2 is a circle, so $\Lambda_M \cong S^1$.

The same openness argument upstairs gives a connected leaf graph Λ_Σ . The action of G on leaves is free. A lifted leaf has endpoints $(x, 0)$ and $(x + t, 1)$. If G^{2n} fixes it, then $2n\beta = 0$. If G^{2n+1} fixes it, then the two endpoint equations give simultaneously $t = (2n + 1)\beta$ and $t = -(2n + 1)\beta$, impossible. Hence $\Lambda_\Sigma \rightarrow \Lambda_M$ is a connected infinite covering of a circle, so $\Lambda_\Sigma \cong \mathbb{R}$.

Orient every lifted leaf from $y = 0$ to $y = 1$. By Lemma 5, its image is a nondegenerate segment, hence determines a continuous oriented support line along Λ_Σ . If ℓ is a leaf, then $G\ell$, reoriented from bottom to top, has the same image segment with the opposite orientation, because G swaps boundary components and $f \circ G = f$. Applying Lemma 6 to the compact interval in Λ_Σ from ℓ to $G\ell$ gives two distinct quotient leaves with perpendicular intersecting support lines. $\square$

Proposition 8 (No $\beta \leq \sqrt{3}$ is realized). *If β is realized, then $\beta > \sqrt{3}$.*

Proof. Let J be a lifted leaf with endpoints $(x, 0)$ and $(x + t, 1)$. Its flat length is $\sqrt{1 + t^2}$. We first show

$$|t| < \beta. \tag{1}$$

If $t = \beta$ or $t = -\beta$, the endpoints are identified by G or G^{-1} , so their images coincide, contradicting Lemma 5. If $t > \beta$, translate so J has endpoints $(0, 0)$, $(t, 1)$. Then GJ has endpoints $(\beta, 1)$, $(t + \beta, 0)$, and the two segments meet at height $(t + \beta)/(2t) \in (0, 1)$. Thus some distinct $p, q \in J$ satisfy $p = Gq$, so $f(p) = f(q)$, contradicting injectivity on the nondegenerate image segment. The case $t < -\beta$ is identical using G^{-1} .

Parametrize the boundary circle of M_β modulo 2β by assigning coordinate u to $(u, 0)$ and coordinate $v - \beta$ to $(v, 1)$. Hence the boundary arcs between the endpoints of a leaf of span t have lengths

$$\beta - t, \quad \beta + t, \quad (2)$$

both positive by (1). Also every nonconstant boundary arc is strictly shortened by F : the quotient boundary is subdivided into finitely many boundary edges, and on each such edge F has constant length factor $\lambda_e < 1$; summing over subsegments gives

$$\ell_F(\gamma) < \ell_0(\gamma). \quad (3)$$

Choose the two leaves T, B from Lemma 7, and let their support lines meet perpendicularly at P . Since distinct leaves have disjoint relative interiors and F is embedded, P cannot lie in the relative interiors of both image segments. Rename so that $F(B)$ lies in a closed ray from P . Let q be the endpoint of B farther from P , and put $Q = F(q)$. Then

$$y := \text{dist}(Q, \text{line } F(T)) = |Q - P| \geq |F(B)| > |B|_0 \geq 1,$$

so in fact

$$y > 1. \quad (4)$$

Lift T with endpoints $(x, 0)$, $(x + t, 1)$. Replacing this lift by its G -translate changes t to $-t$, so it interchanges the two lengths in (2). Choose the lift so that the boundary arc D containing q has length $\beta + t$; let the other arc be H , of length $\beta - t$. Let A, C be the endpoints of $F(T)$, and set

$$L = |A - C| = |F(T)| > \sqrt{1 + t^2}. \quad (5)$$

The arc H connects A to C , so by (3),

$$\beta - t = \ell_0(H) > \ell_F(H) \geq L > \sqrt{1 + t^2},$$

hence

$$\beta > \sqrt{1 + t^2} + t. \quad (6)$$

The arc D connects A to C and passes through Q . Therefore $\ell_F(D) \geq |A - Q| + |Q - C|$. Among points at distance y from the line AC , the sum of distances to A and C is minimized when the projection is the midpoint of AC , giving

$$|A - Q| + |Q - C| \geq \sqrt{L^2 + 4y^2}.$$

Using (4) and (5),

$$\ell_F(D) > \sqrt{5 + t^2}.$$

By (3),

$$\beta + t = \ell_0(D) > \ell_F(D) > \sqrt{5 + t^2},$$

so

$$\beta > \sqrt{5 + t^2} - t. \tag{7}$$

Combining (6) and (7),

$$\beta > \max\{\sqrt{1 + t^2} + t, \sqrt{5 + t^2} - t\}.$$

For every real t , this maximum is at least $\sqrt{3}$. Indeed, if both terms were $< \sqrt{3}$, then squaring $\sqrt{1 + t^2} + t < \sqrt{3}$ gives $t < 1/\sqrt{3}$, while squaring $\sqrt{5 + t^2} - t < \sqrt{3}$ gives $t > 1/\sqrt{3}$, a contradiction. Hence $\beta > \sqrt{3}$. $\square$

2 The upper bound

We quote the classical Sadowsky–Wunderlich construction in the precise form needed.

Theorem 9 (Sadowsky–Wunderlich input). *For every $\mu > \sqrt{3}$, there is a map $E_\mu : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}^3$ such that:*

1. $E_\mu(X + \mu, 1 - Y) = E_\mu(X, Y)$;
2. *the induced quotient Möbius band is embedded;*
3. *the quotient surface is $C^{1,1}$, piecewise C^∞ , and developable;*
4. *it is foliated by full-width straight rulings, and E_μ maps each ruling affinely and isometrically to a segment;*
5. *the boundary lifts are parametrized by arclength in the X -coordinate;*
6. *the construction remains embedded on a slightly wider strip.*

Proof. This is the three-cylinder construction of Sadowsky, with Wunderlich’s analysis of the developable Möbius band; see [1, 2]. In that construction the band consists of three planar pieces and three cylindrical pieces joined along generators with matching tangent planes, and the bending is isometric. The standard separation computation gives maximum half-width

$$b_{\max} = \frac{\mu}{2\sqrt{3}} - \frac{3\pi d}{4}$$

for auxiliary cylinder diameter $d > 0$. Since $\mu > \sqrt{3}$, choosing $0 < d < 2(\mu - \sqrt{3})/(3\pi\sqrt{3})$ gives $b_{\max} > 1/2$, hence a width-one embedded strip, and the strict inequality gives the slightly wider embedded extension. $\square$

For $L, c > 0$, write

$$H_{L,c}(X, Y) = (X + L, c - Y) \quad \text{on } \mathbb{R} \times [0, c].$$

Lemma 10 (Invariant PL approximation). *Let $E : \mathbb{R} \times [0, c] \rightarrow \mathbb{R}^3$ be a scaled Sadowsky band invariant under $H_{L,c}$, with full-width rulings*

$$R_x = [(x, 0), (h(x), c)].$$

Fix x_0 , put $\phi(x) = h(x) + L$, and set

$$S = \max_{x \in [x_0, \phi(x_0)]} |h(x) - x|.$$

For every $D > S$ and every $\varepsilon > 0$, there are an $H_{L,c}$ -invariant clean triangulation $\mathcal{T}_0$ and an $H_{L,c}$ -invariant PL map F_0 such that:

1. $F_0(P) = F_0(Q)$ iff $P = H_{L,c}^k(Q)$ for some $k \in \mathbb{Z}$;
2. every boundary edge is nonexpanded;
3. every transverse edge PQ satisfies

$$|F_0(P) - F_0(Q)| \geq |P - Q| - \varepsilon;$$

4. every transverse edge has horizontal span $< D$.

Proof. The noncrossing full-width rulings imply that h is a strictly increasing homeomorphism. The map $H = H_{L,c}$ sends R_x to $R_{\phi(x)}$, so

$$h(\phi(x)) = x + L, \quad \phi^2(x) = x + 2L. \tag{8}$$

Also

$$|h(x) - x| < L. \tag{9}$$

If $h(x) - x = L$, the endpoints of R_x are identified by H , contradicting the isometric non-degenerate image of R_x . If $h(x) - x > L$, then R_x meets $H(R_x)$ in the interior of the strip; hence two distinct points of R_x have the same image under E , again impossible. The case $h(x) - x < -L$ follows using H^{-1} . Thus $\phi(x) > x$.

Choose $\rho > 0$ with $S + \rho < D$ and $2\rho < \varepsilon$. Choose a partition

$$x_0 < x_1 < \cdots < x_N = \phi(x_0)$$

so fine that $x_{i+1} - x_i < \rho$, $h(x_{i+1}) - h(x_i) < \rho$, all seams of the piecewise smooth Sadowsky band occur among the rulings R_{x_i} , and the embeddedness condition proved below holds. Extend by $x_{i+N} = \phi(x_i)$. Then $x_{i+2N} = x_i + 2L$. Let

$$B_i = (x_i, 0), \quad T_i = (h(x_i), c).$$

The quadrilaterals $B_i B_{i+1} T_{i+1} T_i$ tile the strip. For $0 \leq i < N$, triangulate by the diagonal $B_i T_{i+1}$, and extend H -invariantly. Define F_0 on vertices by $F_0(B_i) = E(B_i)$, $F_0(T_i) = E(T_i)$, and extend affinely on triangles. Then F_0 is H -invariant.

Boundary edges are nonexpanded because their images are chords of boundary arcs parametrized by arclength. A ruling edge is mapped isometrically. For a diagonal $B_i T_{i+1}$, set $d = |B_i - T_{i+1}|$ and $r = |B_i - T_i|$. Since $|T_i - T_{i+1}| < \rho$, one has $|d - r| < \rho$, and

$$|F_0(B_i) - F_0(T_{i+1})| \geq |E(B_i) - E(T_i)| - |E(T_i) - E(T_{i+1})| > r - \rho > d - 2\rho > d - \varepsilon.$$

The other diagonal type is identical, using $x_{i+1} - x_i < \rho$. By (8), the absolute span $|h(x) - x|$ is preserved under ϕ , so all ruling spans are at most S ; diagonal spans are $< S + \rho < D$.

It remains to justify the promised embeddedness for sufficiently fine partition. Let $N = (\mathbb{R} \times [0, c]) / \langle H \rangle$. By Theorem 9, E extends to a slightly wider compact embedded $C^{1,1}$ surface $\widehat{N}$, so there is a tubular neighborhood U with nearest-point projection $\pi : U \rightarrow \widehat{E}(\widehat{N})$. The secant PL maps converge uniformly to E ; because E is affine on rulings, $C^{1,1}$ on each piece, and the seams are included among the ruling edges with matching tangent planes, the maps

$$\Psi = \widehat{E}^{-1} \circ \pi \circ F_0 : N \rightarrow \widehat{N}$$

converge to the inclusion $N \hookrightarrow \widehat{N}$ in C^1 on each piece.

For sufficiently small C^1 -error, Ψ is injective. Choose finitely many coordinate disks with convex coordinate images and $\delta > 0$ so that points of N at distance $< \delta$ lie in a common disk. In such a disk, if the derivative of Ψ differs from the identity by $< 1/2$, then integration along the coordinate segment gives $|\Psi(p) - \Psi(q)| \geq \frac{1}{2}|p - q|$. For pairs with distance $\geq \delta$, compactness and injectivity of the inclusion give a positive minimum separation; sufficiently small C^0 -error preserves this separation. Hence Ψ is injective. If the quotient map induced by F_0 identified two points, then Ψ would identify them, contradiction. Thus the quotient PL map is embedded, which is equivalent to assertion (1) after lifting. $\square$

Proposition 11 (Every $\beta > \sqrt{3}$ is realized). *For every $\beta > \sqrt{3}$, there is a realization of β .*

Proof. Choose μ with $\sqrt{3} < \mu < \beta$, and take the Sadowsky band E_μ . Let $S_\mu = \max |h(x) - x|$ over one fundamental interval of rulings, and choose $D_\mu > S_\mu$. Choose $a < 1$, close to 1, with $a > \mu/\beta$, and set

$$c = \frac{a\beta}{\mu}, \quad L = a\beta.$$

Then $c > 1$ and $L = c\mu$. Scale E_μ by c , giving a width- c , length- L band invariant under $H_{L,c}$. Its span bound scales to cS_μ .

Choose a close enough to 1 so that

$$m := \min_{0 \leq u \leq D_\mu} \left(c\sqrt{1+u^2} - \sqrt{\left(\frac{cu}{a}\right)^2 + 1} \right) > 0. \quad (10)$$

This is possible because at $a = 1$ one has $c = \beta/\mu > 1$, and $c\sqrt{1+u^2} > \sqrt{c^2 u^2 + 1}$ on the compact interval $[0, D_\mu]$. Choose $0 < \varepsilon < m$ and apply Lemma 10 to the scaled band with $D = cD_\mu$. We get $\mathcal{T}_0$ and F_0 .

Define

$$A : \Sigma \rightarrow \mathbb{R} \times [0, c], \quad A(x, y) = (ax, cy).$$

Because $L = a\beta$,

$$A(G_\beta(x, y)) = (ax + a\beta, c - cy) = H_{L,c}(A(x, y)).$$

Thus A conjugates G_β to $H_{L,c}$. Put

$$f = F_0 \circ A, \quad \mathcal{T} = A^{-1}(\mathcal{T}_0).$$

Then $\mathcal{T}$ is a countable locally finite G_β -invariant clean triangulation, and f is affine on every triangle.

If pq is a boundary edge, then

$$|f(p) - f(q)| \leq |Ap - Aq| = a|p - q| < |p - q|.$$

If pq is a transverse edge, let s be the horizontal span of $ApAq$. By Lemma 10, $s < cD_\mu$; write $s = cu$ with $0 \leq u < D_\mu$. Then

$$|Ap - Aq| = c\sqrt{1 + u^2}, \quad |p - q| = \sqrt{\left(\frac{cu}{a}\right)^2 + 1}.$$

Using Lemma 10 and $\varepsilon < m$, (10) gives

$$|f(p) - f(q)| \geq |Ap - Aq| - \varepsilon > |p - q|.$$

Thus f is a squeeze map on every triangle. Finally, for all $p, q \in \Sigma$,

$$f(p) = f(q) \iff F_0(Ap) = F_0(Aq) \iff Ap = H_{L,c}^k(Aq) \text{ for some } k \in \mathbb{Z} \iff p = G_\beta^k(q) \text{ for some } k \in \mathbb{Z}.$$

Therefore β is realized. □

Proof of Theorem 2. Proposition 8 shows that every realized number is $> \sqrt{3}$. Proposition 11 shows that every number $> \sqrt{3}$ is realized. Hence the realized numbers are exactly $(\sqrt{3}, \infty)$. □

Proof of Theorem 1. By Theorem 2, the set of realized numbers is $(\sqrt{3}, \infty)$, whose infimum is $\sqrt{3}$. □

References

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